HOPPER INFORMATION SERIES

AERODYNAMICS

Quadrotor aerodynamics

From rotor thrust to translation, attitude, and the feedback loops that stabilize every command.

NED / FRD

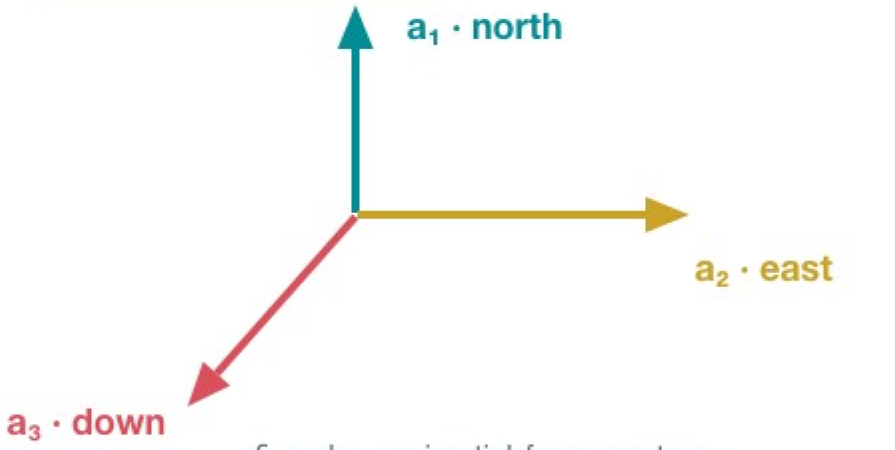
6-DOF MODEL

CONTROL LOOPS

COORDINATE CONVENTION

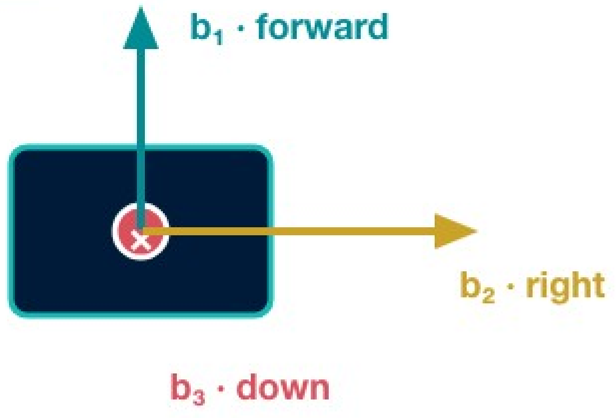
Signs only make sense after the frames are declared

INERTIAL FRAME · NED



ξ_a and v_a are inertial-frame vectors.
Positive z points down.

BODY FRAME · FRD



$R_{a\beta}$ maps body-frame vectors into the inertial frame.

ZYX EULER ANGLES

$$\boldsymbol{\phi} = [\phi_1, \phi_2, \phi_3]^\top = [\text{roll}, \text{pitch}, \text{yaw}]^\top$$

BODY → INERTIAL ROTATION

$$R_{ab} = R_z(\phi_3) R_y(\phi_2) R_x(\phi_1)$$

Rotor speed creates thrust; collective thrust sets height

QUASI-STEADY ROTOR THRUST LAW

$$T_i = c_{ti} u_i, \qquad u_i = \omega_i |\omega_i|$$

POSITIVE TOTAL-THRUST MAGNITUDE

$$F_T = \sum_{i=1}^{4} c_{ti} u_i$$

A TRUE FIRST-ORDER MOTOR MODEL

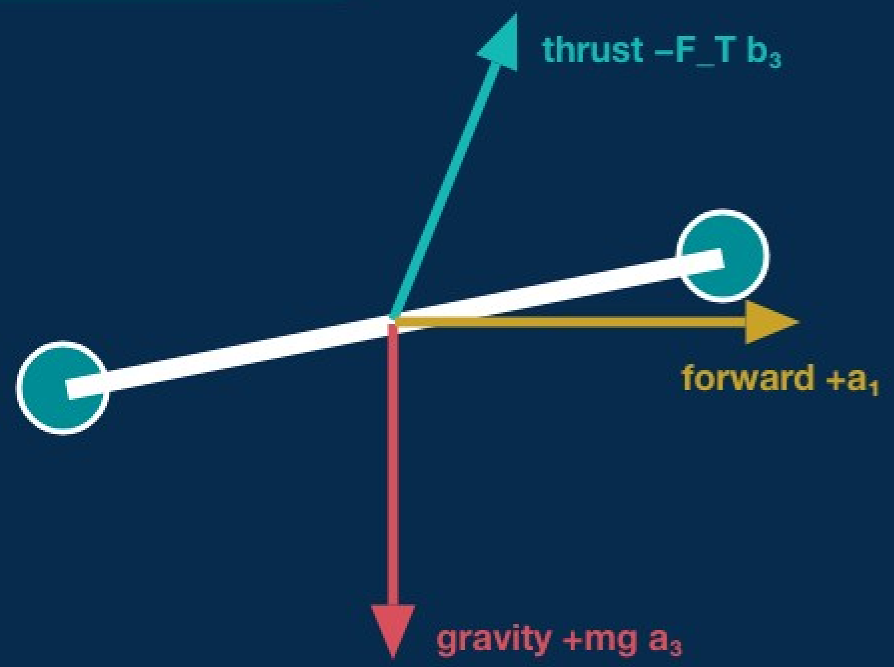
$$\tau_m \dot{\omega}_i + \omega_i = \omega_{i,c}$$

Hover is an attitude-qualified condition

- Level hover: F_T = mg.
- Tilted hover: F_T cos ϕ_1 cos ϕ_2 = mg.
- For one-way rotors, $u_i = \omega_i^2$; the square law is static, not a motor time response.

Students command direction, time, and power; the controller allocates motor commands.

Tilting the body redirects thrust horizontally



Nose-down pitch is $\phi_2 < 0$ in this NED/FRD convention.

FORWARD ACCELERATION · $\phi_3 = 0$

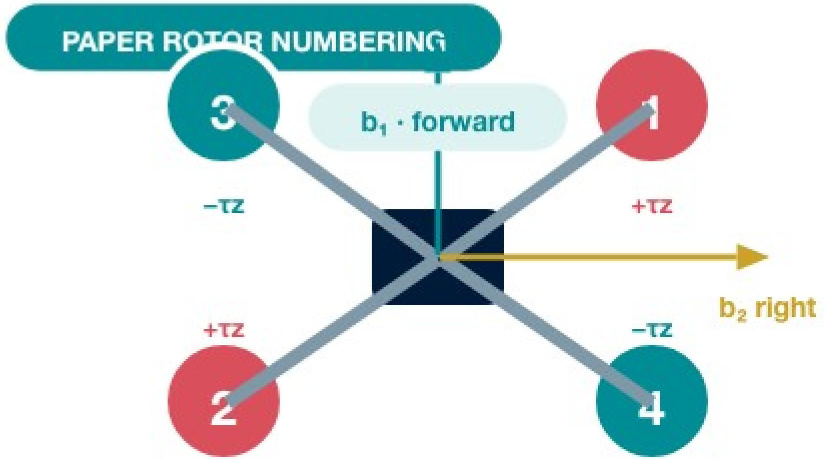
$$\ddot{x} = -\frac{F_T}{m}\sin\phi_2\cos\phi_1 \approx -g\phi_2$$

VERTICAL ACCELERATION · z POSITIVE DOWN

$$\ddot{z} = g - \frac{F_T}{m}\cos\phi_1\cos\phi_2$$

Tilt reduces the vertical thrust component, so altitude hold needs more F_T.

Differential rotor forces create roll, pitch, and yaw moments



SIGN PATTERN

	1	2	3	4
F_T	+	+	+	+
τ_x	−	+	+	−
τ_y	+	−	+	−
τ_z	+	+	−	−

Differential moments are added to the collective command.

THRUST / TORQUE ALLOCATION

$$\begin{bmatrix} F_T \\ \tau_x \\ \tau_y \\ \tau_z \end{bmatrix} = \begin{bmatrix} c_{t1} & c_{t2} & c_{t3} & c_{t4} \\ -d_y c_{t1} & d_y c_{t2} & d_y c_{t3} & -d_y c_{t4} \\ d_x c_{t1} & -d_x c_{t2} & d_x c_{t3} & -d_x c_{t4} \\ c_{d1} & c_{d2} & -c_{d3} & -c_{d4} \end{bmatrix} \mathbf{u}$$

The paper's full 6-DOF model

POSITION KINEMATICS

$$\dot{\boldsymbol{\xi}}_a = \mathbf{v}_a$$

ATTITUDE KINEMATICS

$$\dot{\boldsymbol{\phi}} = J(\boldsymbol{\phi})\boldsymbol{\omega}_b$$

TRANSLATIONAL DYNAMICS

$$m\dot{\mathbf{v}}_a = mg\, a_3 - F_T\, R_{ab}(\boldsymbol{\phi})\, b_3$$

ROTATIONAL DYNAMICS

$$I_m\dot{\boldsymbol{\omega}}_b = \boldsymbol{\tau} - \boldsymbol{\omega}_b \times (I_m\boldsymbol{\omega}_b)$$

v_a: inertial velocity · ω_b = [p,q,r]^T: body rate · I_m: inertia · τ = T u: applied torque

Timed motion is stabilized—but not position-controlled

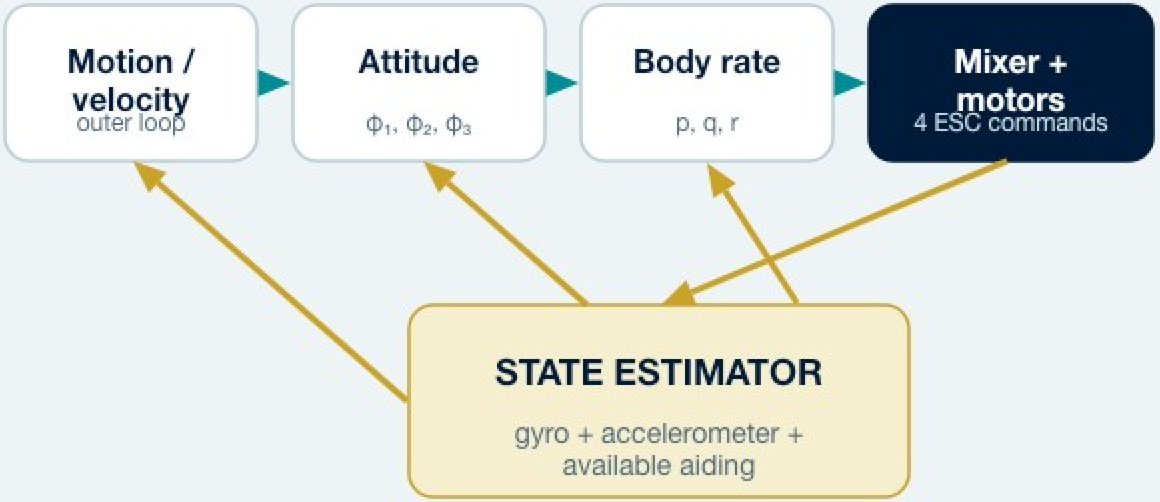
SAFE TIMED MOTION · PROGRAM.JS

```
await drone.takeOff();
try {
  await drone.fly("forward", 2, 15);
  await drone.wait(1);
} finally {
  await drone.land();
}
```

COMMAND CONTRACT

```
await drone.fly(direction, seconds = 0, power = 0)
```

direction — up / down / left / right / forward / backward
seconds — number ≥ 0 s; default 0
power — −100…100%; default 0
returns — Promise<void>; waits for stabilization



What the timed command does—and does not do

The aircraft stabilizes attitude and rate while the command runs. Without external position feedback, duration × power does not guarantee an exact final displacement.

Paper and simulator conventions are different

ATTACHED PAPER · PHYSICAL MODEL

- Inertial NED: z is positive down.
- Body FRD: b_1 forward, b_2 right, b_3 down.
- Full translational and rotational coupling.
- Rotor allocation and gyroscopic cross-product term retained.

SIMULATOR · CLASSROOM APPROXIMATION

- Altitude z is positive up.
- Positive stored pitch means nose-down / forward.
- Horizontal acceleration uses g tan(tilt).
- Attitude and vertical response are simplified and damped.

SIMULATOR POWER → TILT MAPPING

$$\theta_{\text{sim}} = 15^\circ\ \operatorname{sgn}(P) \left(\frac{|P|}{100}\right)^{0.78}$$

3.42°
AT 15% POWER

10°
AT 59.5% POWER

Use the paper for signs and dynamics; use simulation.ts to predict the classroom simulator.